I. Fill in the blanks (points each, in total).

. Turn once about the vertical axis. A ring has free radius , mass and stiffness . The surface is smooth; the ring is elastic, circular and horizontal. For it to remain at rest at any height, the curve equation is

. The frame is a 3D projection of a 4D hypercube. It consists of inner and outer cubes with corresponding vertices linked. If every edge is made of material of resistance , the equivalent resistance across is

. A bee sees light of mean wavelength . Its eye has about ommatidia. They densely cover the whole compound eye. Each intricate unit has a clear round collector (corneal lens) above a clear conical crystalline body. Light focuses on photosensitive cells at the cone tip and cannot cross its walls to another unit. Each makes one image point (pixel); these form a complete image.
Model the compound eye as a spherical cone with spherical radius of . Then the optimum lens diameter is about **(give two significant figures)**

. Open-circuit voltage and short-circuit current are key for a solar cell. For this semiconductor, ; the symbols denote respectively